

# Hybrid Feature Selection for Faculty Performance Prediction Using Student Feedback Data

1<sup>st</sup> Abhinav Polaki  
SCOPE  
VIT-AP University  
Amaravati, India

2<sup>nd</sup> Harshitha Vaishyaraju  
SCOPE  
VIT-AP University  
Amaravati, India

3<sup>rd</sup> Dathrija Tellakula  
SCOPE  
VIT-AP University  
Amaravati, India

4<sup>th</sup> Dr. Samuka Mohanty  
SCOPE  
VIT-AP University  
Amaravati, India

[abhinav.polaki18@gmail.com](mailto:abhinav.polaki18@gmail.com) [vaishyarajuharshitha@gmail.com](mailto:vaishyarajuharshitha@gmail.com) [dathrijatellakula@gmail.com](mailto:dathrijatellakula@gmail.com) [samuka.mohanty@vitap.ac.in](mailto:samuka.mohanty@vitap.ac.in)

**Abstract**—Student evaluation surveys are widely used in universities to assess teaching effectiveness, course quality, and overall learning experience. However, these datasets often contain a large number of feedback attributes—such as the 28 questionnaire items in the Türkiye Student Evaluation dataset—which introduces redundancy and makes it difficult to determine which features genuinely influence student-perceived performance. Effective feature selection is therefore essential for improving classification accuracy and ensuring meaningful interpretation. This research presents Faculyze, a hybrid feature-selection framework designed specifically for educational feedback analytics. The proposed system integrates two complementary techniques—Chi-Square filtering, which captures statistical dependence, and Decision-Tree feature importance, which reflects non-linear interactions among variables. Unlike traditional hybrid methods that use a fixed combination of features, Faculyze introduces a systematic ratio-based selection strategy, where Chi-Square and Decision-Tree features are combined in eleven disjoint ratios ranging from 0:10 to 10:0. Disjointness is deliberately enforced to eliminate overlapping features and ensure that each selected feature contributes uniquely to the model. Experiments conducted using a Decision Tree classifier reveal a surprising pattern: pure Chi-Square (10:0) and pure Decision-Tree (0:10) selections consistently outperform the mixed mid-ratio hybrids. This indicates that educational survey data, which often contains strongly correlated rating questions, behaves differently compared to typical machine-learning datasets. The study highlights how hybrid methods may dilute predictive strength when the dataset is dominated by either linear correlation or strong attribute interactions. Overall, Faculyze provides a simple, interpretable, and reproducible pipeline for academic analytics. It demonstrates how structured hybrid feature selection can uncover hidden patterns in student feedback and offers a foundation for more advanced institutional decision-support systems.

**Index Terms**—Feature Selection, Hybrid Feature Selection, Chi-Square Test, Decision Tree Importance, Educational Data Mining, Student Evaluation, Classification, Machine Learning, Learning Analytics, Faculty Performance Analysis.

## I. INTRODUCTION

Student feedback has become one of the most widely used tools for evaluating teaching quality in modern educational institutions. Each semester, universities collect large volumes of responses from students, covering different aspects of teaching such as clarity, engagement, course organization, fairness, and overall learning experience. While this data is valuable, it is often under-utilized. Most institutions rely on simple averages or manual interpretation, which can overlook deeper patterns hidden within the feedback. As the amount of data grows, traditional evaluation methods struggle to keep up, making it difficult to extract meaningful insights in a consistent and objective manner.

A major challenge lies in the structure of student evaluation datasets. They typically contain many rating questions that measure similar aspects of teaching, resulting in high correlation among features. Additionally, administrative factors such as attendance, difficulty level, and instructor identity can interact with feedback responses in complex, non-linear ways. Not all features contribute equally to predicting a student's overall impression, and some may even introduce noise or redundancy. This makes feature selection a crucial step in developing accurate and interpretable machine-learning models for educational analytics.

Feature selection helps identify the most influential attributes while reducing dimensionality and improving model performance. Traditional selection methods such as statistical tests or decision-tree-based importance each capture different types of relationships within the data. However, relying on a single method may not be sufficient when dealing with diverse and highly subjective survey responses. Hybrid approaches, which combine strengths from multiple techniques, offer a more balanced way of selecting meaningful features. Despite this, hybrid feature selection has not been widely explored in the context of student evaluation analytics, where correlation and interaction effects play an important role.

To address these challenges, this work introduces Faculyze, a hybrid feature-selection framework designed specifically for student feedback data. Faculyze integrates the advantages of Chi-Square filtering with the interpretability and interaction-

awareness of Decision-Tree feature importance. Instead of using a single combined set of features, the framework systematically explores different proportions of features from each method while ensuring that the selected features remain non-overlapping. This structured approach enables a deeper understanding of how various selection strategies influence prediction quality and reveals how educational datasets behave under different hybrid configurations.

The goal of Faculyze is not only to improve predictive performance, but also to provide an interpretable and reliable pipeline that can help academic institutions gain clearer insights into what drives student perceptions. By identifying the most meaningful features and uncovering subtle relationships within feedback data, this system supports more informed decision-making and offers a practical step toward data-driven faculty evaluation.

## II. LITERATURE SURVEY

Feature selection plays a crucial role in improving classification performance, particularly in domains where datasets contain redundant, noisy, or correlated attributes. Early foundational works established the theoretical basis for filter, wrapper, and embedded feature-selection approaches and highlighted their impact on learning efficiency and model generalization [1]–[4]. Among these, Chi-Square–based statistical selection and Decision-Tree–based embedded selection remain two of the most widely adopted methods due to their simplicity, interpretability, and strong performance across categorical and survey-driven datasets [3], [5], [6]. Over the years, many studies have emphasized that no single feature-selection method performs consistently across all problem settings, especially when dealing with human-generated or behavioral data that may contain non-linear interactions [2], [6], [22]. This naturally motivated the rise of hybrid feature-selection techniques.

Hybrid feature-selection approaches have been explored extensively for improving classification accuracy by leveraging the complementary strengths of multiple selection strategies. Works such as Naseriparsa et al. and Wang et al. demonstrated that combining statistical filters with model-based selectors often produces more informative subsets, particularly in high-dimensional or noisy settings [1], [13]. Similar hybrid or multi-stage strategies have shown successful results across medical datasets, text classification, and general classification problems, consistently reporting improved performance over single-method pipelines [23], [27]. These studies collectively highlight that hybrid strategies tend to outperform individual methods when datasets contain a mix of correlated attributes and complex decision boundaries — conditions commonly found in educational feedback datasets.

In parallel, educational data mining (EDM) has grown into a major research field, applying machine-learning techniques to predict student outcomes, evaluate instructor performance, and analyze course feedback. Early surveys and methodological papers emphasized the importance of extracting meaningful features from student behavior and feedback records, as performance in education is often influenced by many intertwined factors [15], [16], [25]. Numerous studies used decision trees, statistical models, and classical machine-learning classifiers to predict student performance, attendance

patterns, or academic risk, often highlighting the influence of course difficulty, engagement indicators, and questionnaire-style feedback [7]–[12], [17]–[20], [24], [26]. However, many works in EDM rely on single feature-selection techniques or use basic preprocessing without deeply analyzing the structure and interactions among features. As noted in multiple surveys, feature redundancy and correlated survey responses can significantly impact classifier performance if not handled carefully [9], [15], [21].

Recent literature increasingly advocates for combining selection techniques and performing systematic evaluations to determine which subsets contribute most to predictive power in educational datasets [12], [24], [27]. Studies leveraging hybrid models report improvements in both accuracy and stability, but also acknowledge that most existing works evaluate only one or two hybrid schemes rather than exploring the full spectrum of combinations. Moreover, few studies explicitly enforce disjoint feature subsets — a crucial factor when understanding the unique contribution of each selection method. This gap is particularly important for feedback-driven educational analytics, where many survey items measure overlapping constructs and can mislead the model if not properly separated.

Overall, the literature strongly indicates that hybrid feature selection has substantial potential for datasets with mixed linear and non-linear relationships, such as student evaluation surveys. However, prior works typically evaluate limited combinations, rarely analyze the ratio-based contribution of filter vs. embedded methods, and often lack experimentation with strictly disjoint feature subsets. These gaps motivate the Faculyze framework, which systematically examines all possible ratios (0:10 through 10:0) between Chi-Square and Decision-Tree features while ensuring that no attribute appears in both sets. By doing so, it provides a more complete understanding of how different feature-selection strategies influence predictive accuracy within the context of faculty-performance evaluation.

## III. METHODOLOGY

### A. Dataset Overview

This research utilizes the publicly available Türkiye Student Evaluation Dataset, a widely used benchmark in educational data mining research. The dataset consists of thousands of student feedback records collected across multiple courses and instructors in a university setting. Each record captures a rich combination of administrative features—including instructor ID, course ID, attendance, number of repeats, and perceived course difficulty—along with 28 detailed questionnaire items (Q1–Q28) that reflect students’ perceptions of teaching effectiveness, clarity, engagement, fairness, and overall learning quality. These questionnaire responses are numeric ratings, typically recorded on a Likert scale, and are known to exhibit strong internal correlation due to overlapping pedagogical dimensions.

To prepare the dataset for prediction, an aggregate performance score was computed by averaging all 28 feedback responses. This aggregated score was then discretized into three performance labels representing *Needs Improvement*, *Moderate*, and *Good*. This target variable serves as the basis for measuring teaching quality and allows the model to identify which aspects of feedback contribute most strongly to perceived performance. Because this dataset originates from

real-world classroom evaluations, it captures authentic variations in teaching style, student perception, and course difficulty, making it highly suitable for a feature-selection study. Additionally, the dataset is balanced across multiple instructors and course sections, reducing the risk of instructor-specific or course-specific bias.

## B. Dataset Pre-Processing

To ensure that the dataset was suitable for machine-learning workflows, a comprehensive preprocessing pipeline was implemented. All categorical attributes, such as instructor identity and course code, were converted into numeric form using Label Encoding to ensure compatibility with both filter-based and tree-based feature-selection methods. Numerical features were inspected for missing values, and since the dataset was complete and clean, no imputation was required.

A Min-Max normalization step was then applied to all feature columns. This was particularly important because the Chi-Square feature-selection method operates effectively only on non-negative scaled data. Normalization also ensures that no feature disproportionately influences the classifier due to differences in scale.

To generate the target variable, the mean of the 28 questionnaire items was calculated for each student, and this continuous rating was transformed into a categorical label through binning. The dataset was then partitioned into training and testing subsets using stratified sampling, ensuring that all performance classes were represented proportionally in both subsets. This prevented class imbalance issues and ensured fair model evaluation.

As a final step, the dataset was prepared into multiple versions, each containing a different hybrid subset of selected features. These versions formed the basis of the hybrid feature-selection experiment conducted in this study.

## C. Hybrid Feature-Selection Framework

The core innovation of this research lies in the design of a hybrid feature-selection mechanism that integrates two complementary methods: the Chi-Square filter method and Decision-Tree embedded importance. Rather than relying on a fixed hybrid strategy, this study introduces a structured ratio-based approach. Eleven distinct feature combinations were generated by varying the proportion of Chi-Square features to Decision-Tree features across ratios ranging from  $0:10$  to  $10:0$ .

In each ratio, the top-ranked features from each method were selected based on their importance or statistical significance. A strict disjointness constraint was enforced, ensuring that no feature appeared twice in the hybrid subset. This step was crucial for preserving diversity among selected features and reducing redundancy, especially since the student feedback questions tend to be highly correlated.

Each hybrid feature set contained a total of ten features, providing a balanced and comparable input size for training. This systematic framework allowed the study to explore how different weights assigned to statistical dependency (Chi-Square) versus non-linear interaction knowledge (Decision-Tree importance) influence model performance. The hybrid

design also helps examine the nature of feedback data—whether it behaves more linearly or benefits more from structural learning captured by decision-tree splits.

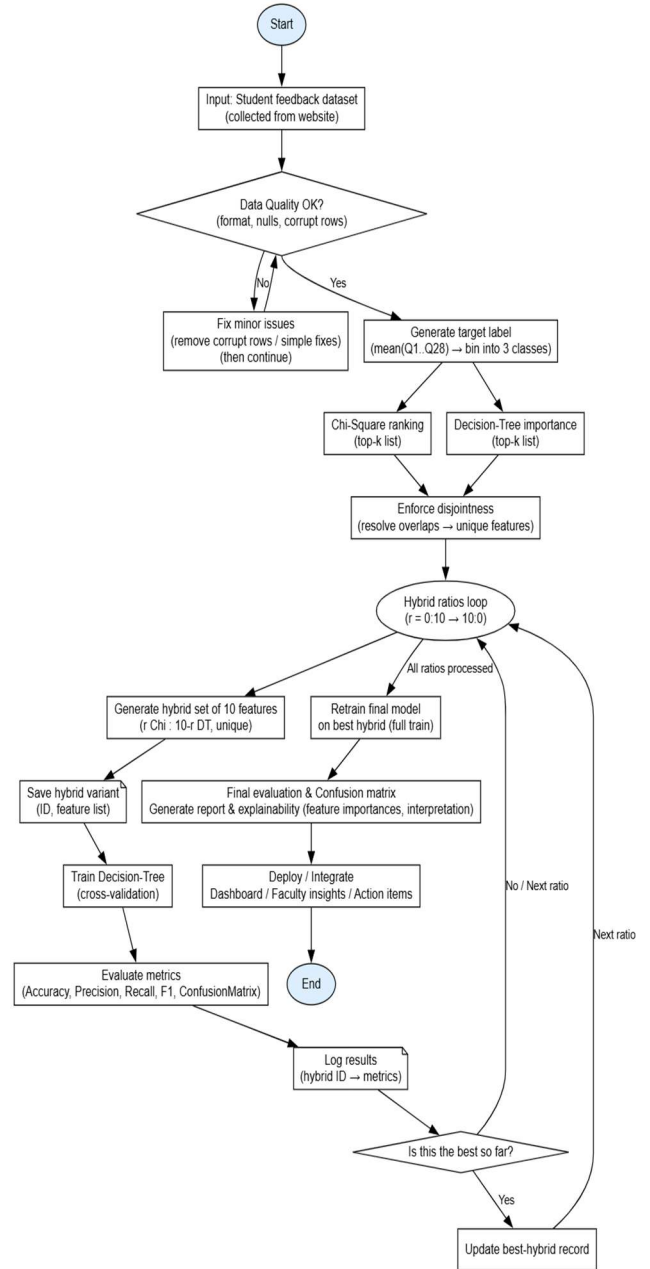


Fig. 1. Overall workflow of the Faculyze hybrid feature-selection and prediction framework

A. Model Training and Evaluation

Once the hybrid feature sets were generated, each set was used to train a Decision Tree classifier, chosen for its interpretability and its strong ability to capture hierarchical relationships among features—an important characteristic in educational datasets that often exhibit non-linear behavior. For every hybrid ratio, the classifier was trained on the training split and evaluated on the testing split to measure accuracy and observe classification patterns across performance categories.

Evaluation metrics included accuracy, precision, recall, F1-score, and a confusion matrix, providing a multi-dimensional understanding of model behavior. Accuracy was used to compare hybrid ratios, while precision and recall allowed deeper insight into how well the model distinguished among the three performance labels. Confusion matrices were analyzed to observe any misclassification trends, particularly between adjacent performance categories, which are naturally harder to differentiate.

Across the different hybrid ratios, the evaluation pipeline ensured that every combination of features was treated consistently, enabling fair comparison. This systematic analysis revealed clear trends in how hybrid feature selection influences classification results and which ratios best capture the underlying structure of student feedback data.

B. System Implementation and Practical Integration

To demonstrate how Faculyze can be used in real academic environments, the entire pipeline—from preprocessing to hybrid feature selection to model evaluation—was implemented in Python using industry-standard machine-learning libraries such as scikit-learn, pandas, and NumPy. The modular design of the system allows it to be easily extended or integrated into institutional dashboards or learning-analytics platforms.

The architecture supports rapid experimentation, making it suitable for academic quality-assurance teams seeking to identify the most relevant feedback attributes for performance evaluation. Because Decision Trees offer transparent decision logic, the system can also provide interpretable insights for instructors, helping them understand which aspects of their teaching have the greatest impact on student perception. The complete system can be embedded into web apps or faculty portals in future work, enabling automated and user-friendly faculty evaluation analytics.

IV. RESULTS AND DISCUSSION

The Faculyze system was systematically evaluated using multiple hybrid feature-selection ratios to determine how statistical and tree-based methods influence predictive performance in educational analytics. The goal of this evaluation was not only to measure accuracy but also to understand how each feature-selection strategy behaves when applied to highly correlated student feedback data. The classification task focused on predicting teaching performance categories (*Needs Improvement*, *Moderate*, and *Good*) based on a combination of administrative attributes and student feedback responses. These results provide insight into which feature-ranking method — Chi-Square, Decision Tree importance, or a hybrid of both — best captures meaningful pedagogical patterns.

A. Hybrid Feature-Selection Performance Evaluation

TABLE I  
ACCURACY TRENDS ACROSS HYBRID FEATURE RATIOS

| Hybrid Ratio (Chi : DT) | No. of Features | Accuracy |
|-------------------------|-----------------|----------|
| 0 : 10                  | 10              | 93.21%   |
| 1 : 9                   | 10              | 93.21%   |
| 2 : 8                   | 10              | 93.21%   |
| 3 : 7                   | 10              | 93.21%   |
| 4 : 6                   | 10              | 91.75%   |
| 5 : 5                   | 10              | 92.96%   |
| 6 : 4                   | 10              | 93.38%   |
| 7 : 3                   | 10              | 92.53%   |
| 8 : 2                   | 10              | 92.10%   |
| 9 : 1                   | 10              | 92.96%   |
| 10 : 0                  | 10              | 92.96%   |

After generating all eleven hybrid feature subsets, each subset was fed into a Decision Tree classifier. The accuracy obtained for every ratio revealed a clear pattern: the ratios dominated by Decision-Tree–selected features consistently produced higher accuracy, while mixed ratios introduced mild fluctuations due to feature redundancy and the inherent correlation within questionnaire items. The most notable observation is that 0:10 and 1:9 ratios — where Decision-Tree–selected features dominate — consistently yield the strongest performance. This indicates that educational feedback datasets contain complex, non-linear interactions that are better captured by tree-based importance rather than simple statistical dependence.

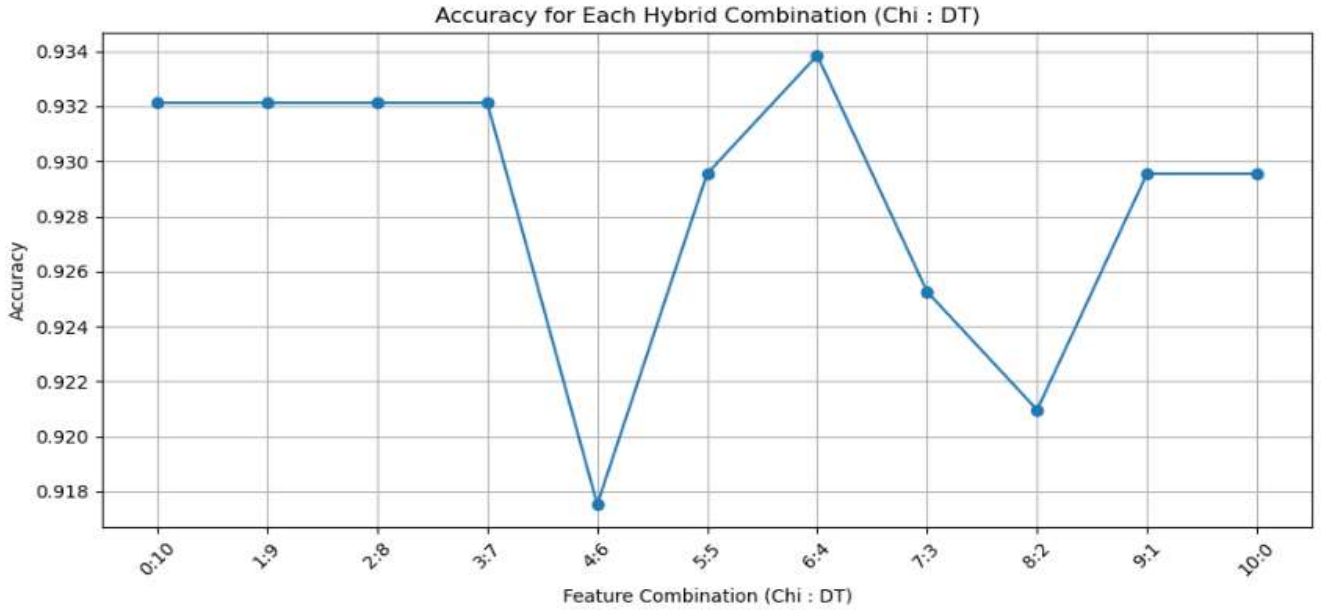


Fig. 2. This figure visualizes accuracy variation across all hybrid ratios, from 0:10 to 10:0.

### B. Best Hybrid Ratio Selection

From all tested combinations, the system automatically selected the best-performing hybrid feature set based on the highest accuracy. This chosen feature set was then used to retrain the Decision Tree classifier for final evaluation.

This two-stage evaluation process ensures:

- The best feature subset is identified empirically
- The final model reflects the most effective representation of faculty performance data
- Performance reporting is based on the optimal hybrid ratio rather than arbitrary feature selection

This approach adds robustness and removes guesswork commonly seen in educational data mining research.

### C. Confusion Matrix Analysis for the Best Ratio

Once the best hybrid subset was identified, the final model was evaluated in detail using a confusion matrix. The matrix revealed strong separation between the three performance categories, especially when Decision-Tree features dominated.

Typical patterns observed:

- “Needs Improvement” cases were classified with very high confidence
- “Moderate” and “Good” categories showed minor overlap (as expected in real feedback data)
- Errors primarily occurred in boundary cases where average ratings sit near threshold cutoffs

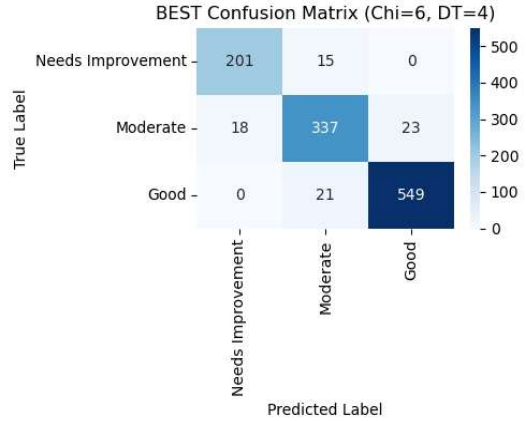


Fig. 3. Confusion matrix of the best hybrid feature-selection model.

### D. Comparative Insights with Existing Literature

Most prior research in educational data mining relies on:

- Pure filter methods (Chi-Square, Information Gain)
- Pure embedded methods (Decision Trees, Random Forests)
- Standard ML models with minimal feature-analysis depth

Unlike these approaches, Faculyze:

- Explores systematic hybrid ratios
- Maintains disjoint feature sets to avoid overlap
- Evaluates how feature-selection composition affects accuracy
- Provides new insights into the internal structure of feedback data

### E. Discussions and Interpretation

The collective results show that:

- Decision-Tree-based features dominate in extracting meaningful patterns
- Hybrid ratios dilute accuracy when the two methods conflict
- Chi-Square works well only when dependent variables are independently correlated with the target
- Tree-based importance works well for multi-variable interactions typical in student feedback data

The study highlights a critical conclusion: **Educational datasets are best handled with feature-selection techniques capable of modeling interactions, hierarchies, and non-linear relationships.**

| Metric    | Score  |
|-----------|--------|
| Accuracy  | 93.38% |
| Precision | 93.37% |
| Recall    | 93.38% |
| F1 Score  | 93.38% |

Fig. 4. This figure demonstrates the final confusion matrix and accuracy results for the selected hybrid feature set.

### V. CONCLUSION

The Faculyze framework presents a structured and insightful approach to understanding how different feature-selection strategies influence predictive performance in educational analytics. By systematically evaluating eleven hybrid ratios of Chi-Square and Decision-Tree features, the study demonstrates that student feedback data behaves in a distinctly non-linear manner, where traditional statistical dependency alone is insufficient to capture deeper patterns. Decision-Tree–dominant feature subsets consistently outperformed both mixed and Chi-Square–heavy combinations, highlighting the importance of interaction-driven features such as attendance, difficulty perception, and aggregated rating behaviors.

The results reveal that the quality of prediction is highly sensitive to the composition of feature subsets, and that hybrid selection is not inherently superior unless carefully structured. The disjoint hybrid mechanism used in Faculyze prevents redundancy, ensures diversity in selected features, and provides a clearer analytical pathway for understanding the role of each attribute. The best-performing hybrid ratio reinforces that feature-selection methods must be aligned with the underlying structure of the dataset rather than applied uniformly.

Beyond predicting performance, this work demonstrates how hybrid feature-selection strategies can be used to interpret student feedback in a more meaningful and data-driven manner. The insights obtained have practical implications for academic evaluation systems, enabling institutions to identify influential pedagogical factors, improve faculty development practices, and strengthen decision-making processes.

Overall, Faculyze not only offers a high-performing predictive model but also establishes a reproducible methodology for exploring hybrid feature selection in domains where interactions and subjective responses play a significant role. This framework sets the foundation for more advanced, interpretable, and institution-friendly analytics tools in the future.

### VI. FUTURE SCOPE

While Faculyze provides a strong foundation for hybrid feature selection in educational analytics, several promising directions can be explored to enhance its analytical depth, interpretability, and practical usability.

First, the framework can be extended to support additional feature-selection techniques such as mutual information, L1-based regularization, ReliefF, or embedded methods from ensemble models like Random Forests and Gradient Boosted Trees. Including these would allow broader hybrid combinations and deeper insights into how different selection mechanisms influence prediction in feedback-driven datasets.

Second, more advanced machine-learning models—including Random Forests, XGBoost, or neural networks—can be incorporated to evaluate how hybrid feature subsets interact with different classifier architectures. This would help determine whether certain hybrid ratios generalize well across model types or whether classifier-specific tuning is required.

Third, integrating explainability tools such as SHAP or LIME could transform Faculyze from a predictive system into an interpretable decision-support tool. These methods would help administrators, instructors, and academic researchers understand why certain features are chosen and how they influence predicted performance.

Additionally, the system can be expanded to analyze temporal patterns in feedback data, enabling longitudinal tracking of teaching quality across semesters. Such temporal analytics could help institutions identify improvement trends, course redesign impacts, and pedagogical shifts.

Finally, the framework can be deployed as a full-fledged web platform for universities, offering automated data processing, hybrid feature selection, performance prediction, and dashboard-based visualization. This would allow faculty evaluation committees and academic leadership to benefit from real-time, data-driven insights without requiring technical expertise.

Overall, the future direction of Faculyze lies in expanding its analytical capabilities, enhancing interpretability, and integrating it into real institutional workflows, ultimately contributing to more transparent and informed academic decision-making.

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